32-bit Microcontrollers Starter Kit

Quick Guide

Version 1.08

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1. Introduction

This Quick Start Guide provides information on using the ABOV 32-bit Cortex-Mx starter kit.

Using the starter kit, you can test designs and functions of ABOV 32-bit Cortex-Mx MCUs. A starter kit board contains a CMSIS-DAP debugger called “A-Link,” which eliminates the need for additional devices for MCU programming and debugging.

In this Quick Start Guide, we will discuss how to operate LED_Blinky, as an example, step by step.

2. User requirements

2.1 Hardware

1. ABOV 32-bit Starter Kit Boards
   A. This Starter Kit is available for the devices based on ARM Cortex-M processor.

Table 1. Lists of ABOV 32-bit Starter Kit

32-bit Series	Starter Kit Name
A31G11x Series	Starter Kit_A31G112CL
A31G12x Series	Starter Kit_A31G123ML
A31G21x Series	Starter Kit_A31G213CL2N
A31G31x Series	Starter Kit_A31G314MMN
A33G52x Series	Starter Kit_A33G527VQN Starter Kit_A33G526RLN
AC30M1x64 Series	Starter Kit_AC30M1464
AC33Mx064 Series	Starter Kit_AC33M4064LBN
AC33Mx128 Series	Starter Kit_AC33M8128L
A33M41x Series	Starter Kit_A34M418VLN STK_A34M418-YLN-A
A31G22x Series	STK-A31G226-ML2N-A
A31G32x Series	STK-A31G324-RLN-A
A31L12x Series	STK-A31L123-RLN-A
A31L21x Series	STK-A31L214-ML2N-A
A33M11x Series	STK-A33M116-RLN-A
A31R71x Series	STK-A31R713-MLN-A

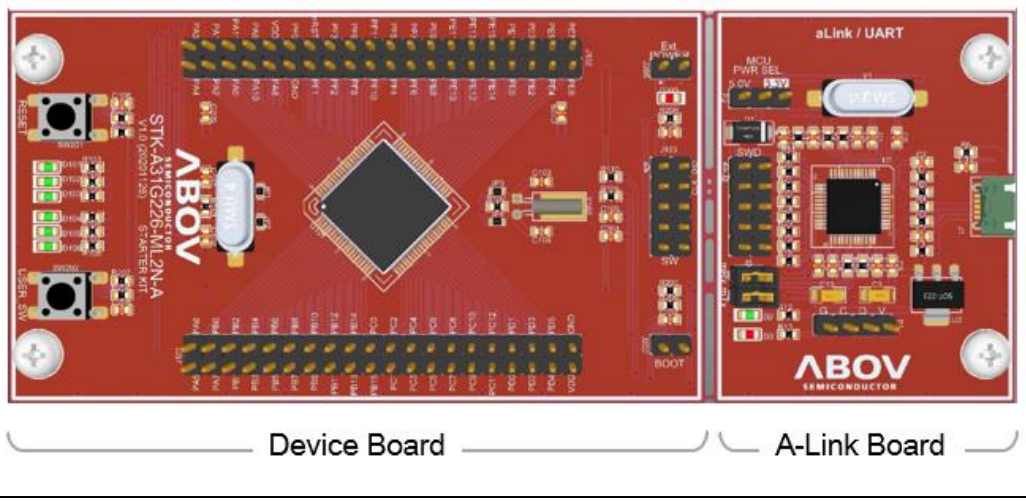


Figure 1. 32-bit Starter Kit Board (Hardware) Example

2.2 Software

1. Keil complier (uVision5)
2. Keil MDK5 software pack (including LED_Blinky sample code) or Example Code software

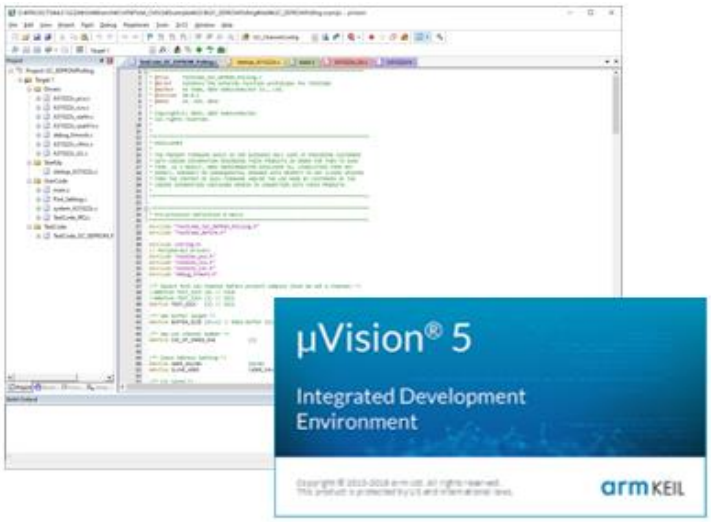


Figure 2. Compiler (Software)

2.3 Reference documents

1. 32-bit Starter Kit Quick Guide
2. Starter Kit HW Manual
3. Starter Kit Schematic
4. Product User's Manual
5. Example Code

■ Documents

Category	Title	Type	Language	Version	Issue Date	Size	Download
Datasheet	DS_A31G22x_Eng		Eng	1.01	2021-01-11	3.3 MB	
User's Manual	UM_A31G22x_Eng		Eng	1.01	2021-01-11	11.1 MB	
Starter Kit	STK 32-bit Starter_Kit_Quick_Guide_Eng		Eng	1.04	2020-12-22	1.7 MB	
Starter Kit	STK_Gerber_A31G226MLN		-	1.1.0	2020-12-22	757 KB	
Starter Kit	STK_Manual_A31G226MLN_Eng		Eng	1.01	2021-01-18	923 KB	
Starter Kit	STK_Schematic_A31G226MLN		-	1.1.0	2020-12-22	1.9 MB	
Example Code	Example_Code_A31G22x		-	1.00	2021-01-04	12.6 MB	
Application Note	AN_Hardware Design Guide for Noise Immunity for 8-bit 32-bit Devices_Eng		Eng	1.00	2021-01-14	1.3 MB	
Application Note	AN_IAR Project Setting Guide_A31G22x_Eng		Eng	1.01	2021-01-18	1.5 MB	
Application Note	AN_KEIL Project Setting Guide_A31G22x_Eng		Eng	1.01	2021-01-18	1.1 MB	
Application Note	AN_ClockSettingGuide_A31G22x_Eng		Eng	1.01	2021-01-18	816 KB	
Application Note	AN_InternalDataFlashMemoryGuide_A31G22x_Eng		Eng	1.00	2020-12-22	948 KB	
Application Note	AN_LEDCircuitDesignGuide_A31G22x_Eng		Eng	1.00	2020-12-22	1.1 MB	
Application Note	AN_OperatingModesandWake-upfunctionGuide_A31G22x_Eng		Eng	1.00	2020-12-22	529 KB	

Figure 3. Reference Documents

2.4 System requirements

1. Windows PC (7, 8, 10)
2. Starter Kit uses Micro-B 5-pin USB Cable.



Figure 4. Window PC & Micro-B 5-pin USB cable

2.5 ABOV website

For detailed information about corresponding software and documents, you can visit our website at https://www.abovsemi.com.

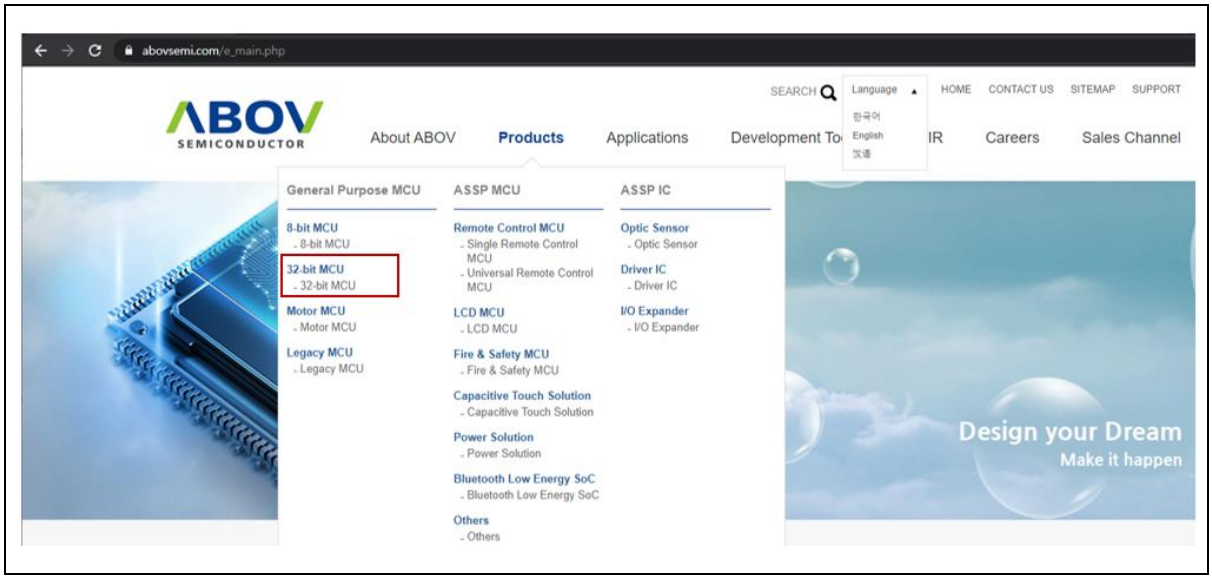


Figure 5. ABOV Semiconductor Website

3. Example: How to operate LED_Blinky

Running application code (LED_Blinky) makes it easier to start the Starter Kit. When following the steps below, LEDs on the board turn on and off:

Step 1. Prepare the Starter kit Board

Step 2. Set up the Starter kit Board

Step 3. Connect the Starter kit to your PC

Step 4. Run the LED_Blinky Project

Step 5. Compiling LED_Blinky

Step 6. Download LED_Blinky

Step 7. Debugging LED_Blinky

Step 8. Check the LED_Blinky Operation

3.1 Prepare the Starter Kit board

Hardware components of the Starter Kit

1. The Starter Kit consists of a device board and an A-Link (CMSIS-DAP) debugger board. It is possible to use separately as needed.
2. Device board configuration
   A. ABOV 32-bit MCU, X-TAL (optional)
   B. Pin Headers connected to MCU
   C. LEDs and switch to check input/output
   D. Reset switch, boot mode selection jumper, debugger connector
3. A-Link (DMSIS-DAP/UART) board configuration
   A. Programmable and debuggable A-Link connected to PC via USB from ABOV 32-bit MCU.
   B. Supports UART Tx/Rx channel.

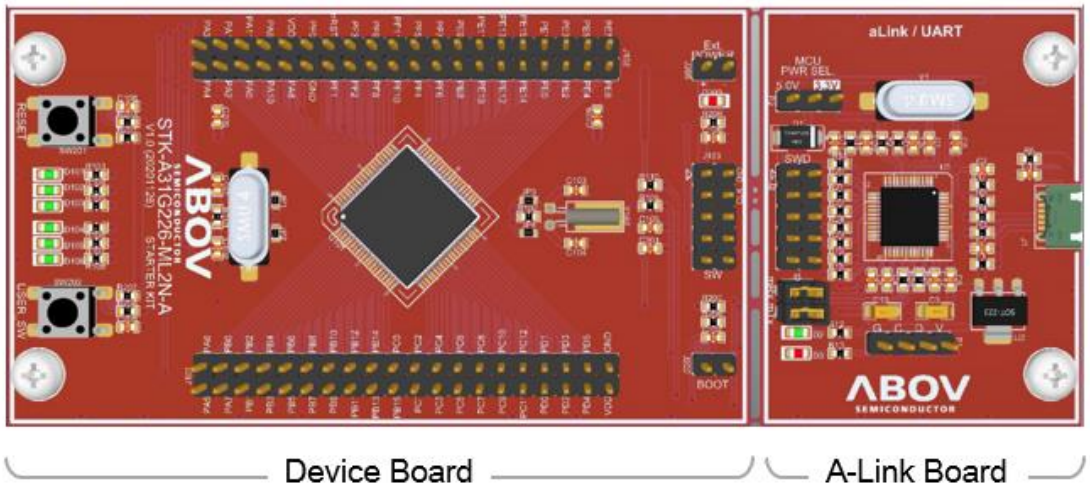


Figure 6. Starter Kit Board

3.2 Set up the Starter Kit Board

Set jumpers to control the Starter Kit

① The Starter Kit uses USB power

Choose 3.3V and 5.0V (check the maximum operation voltage by referring to the specification sheet).

② Boot mode

Open: User program operating mode (Default)

Short: Boot mode (Debugger always enabled)

③ USB to UART Connector

Jumper to use UART through the USB of the A-Link board

Used to connect UART Rx and Tx

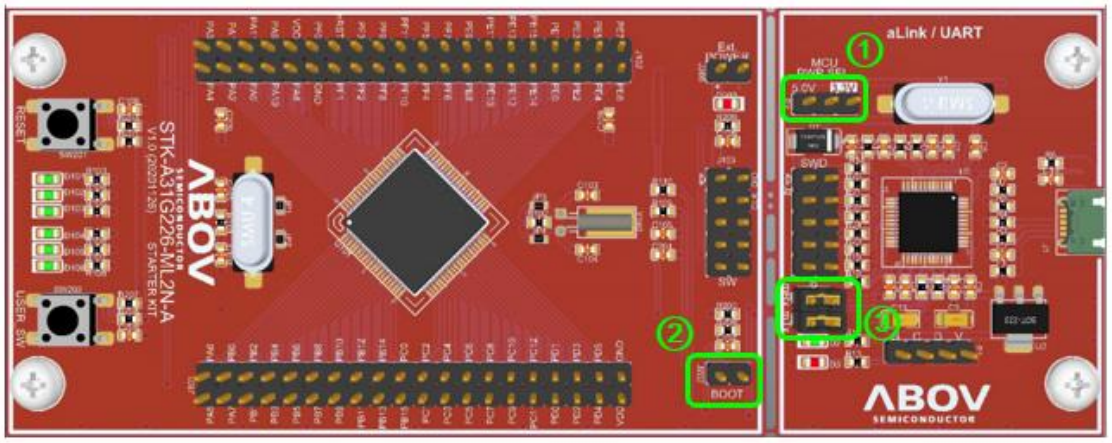


Figure 7. Starter Kit Jumper & Switch

3.3 Connect the Starter Kit to your PC

PC connection via USB on the Starter Kit

Connect the USB 2.0 (micro-B type) cable to the Starter Kit as shown in Figure 8.

① If the PC is properly connected, the power LED on the device board will be turned on.

② The status LED on the A-Link board (debugger) will be turned on.

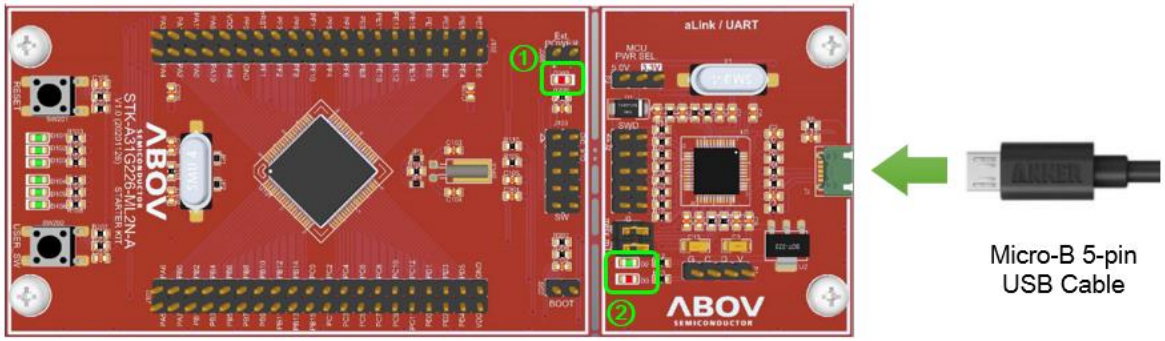


Figure 8. Starter Kit and USB Connection

Recognition of the hardware (device drivers) on your PC

The A-Link board (debugger) uses a USB interface. Once the A-Link board (debugger) is connected, it will be shown as “USB Composite Device”. A-Link does not require driver installation on Microsoft Windows.

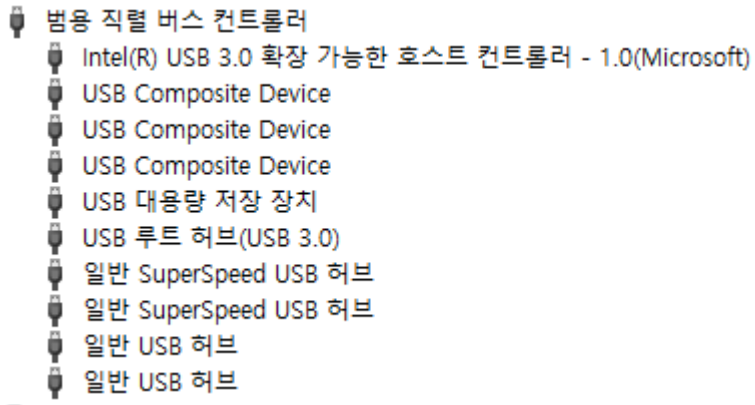
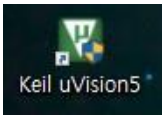


Figure 9. Serial Bus Controller Lists at Device Manager

3.4 Run the LED_Blinky project

Before downloading the LED blinky example, Keil uVision5 S/W must be installed on your PC.

1. Execute “Keil uVision5”.

   ① 

2. Click the Pack Installer icon to run the installer. Pack Installer is a utility to install/update software packs.

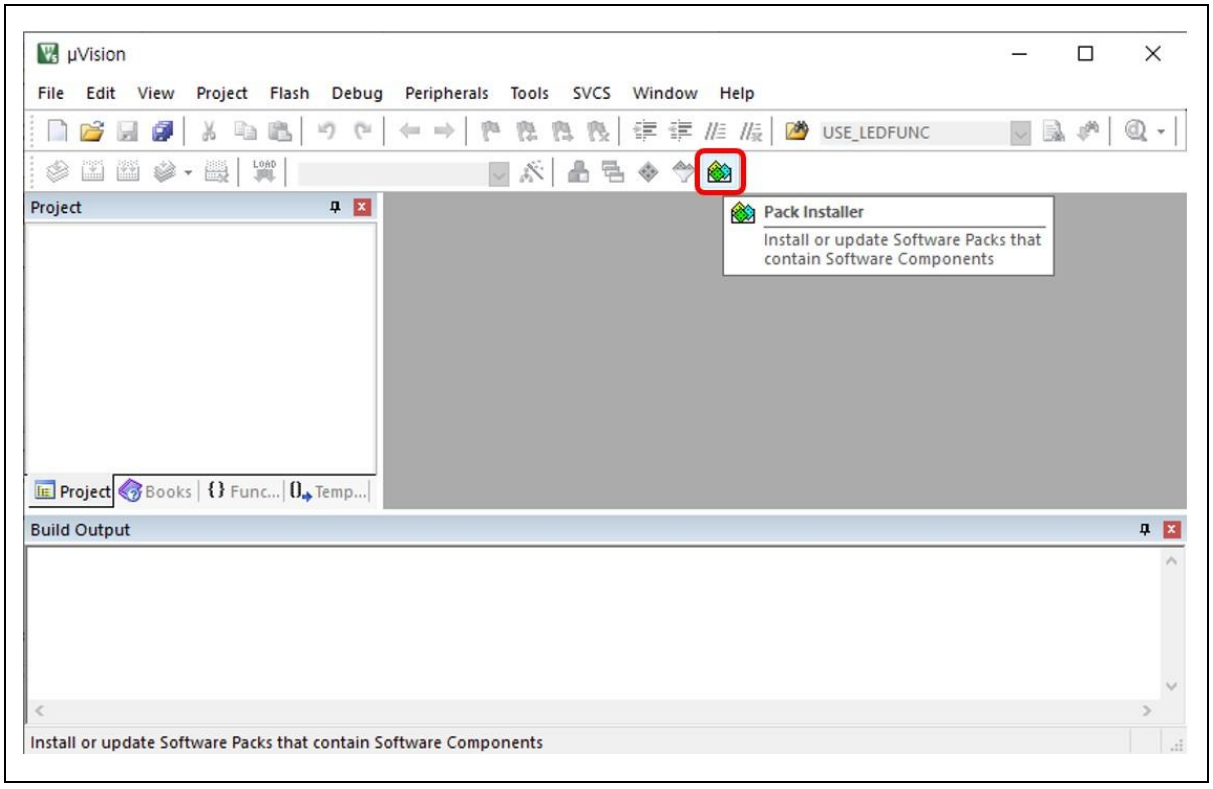


Figure 10. Execute Keil uVision5 and Pack Installer Icon

3. Select the starter kit device and example in Pack Installer.

① Select ABOV Semiconductor in All Devices of Devices tab and Choose Series and target device. (Select the same MCU on the starter kit).

② Select a example to test on the target. StarterKit GPIO Blinky Example in Examples tab.

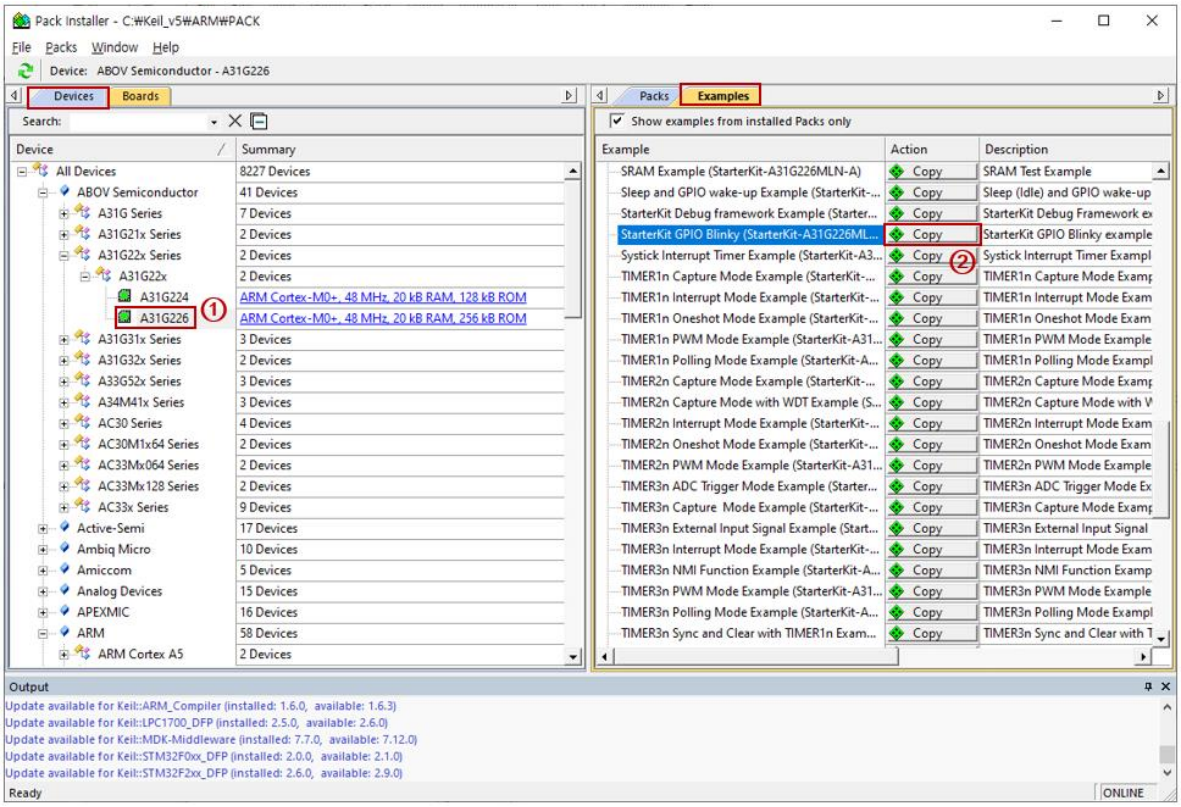


Figure 11. Target Device and Example selection in Pack Installer

4. Create a starter kit example in Pack Installer.

① When the selected example is copied, a dialog box pops up to prompt you to choose a destination folder.

② Select a destination folder and click OK to run uVision5.

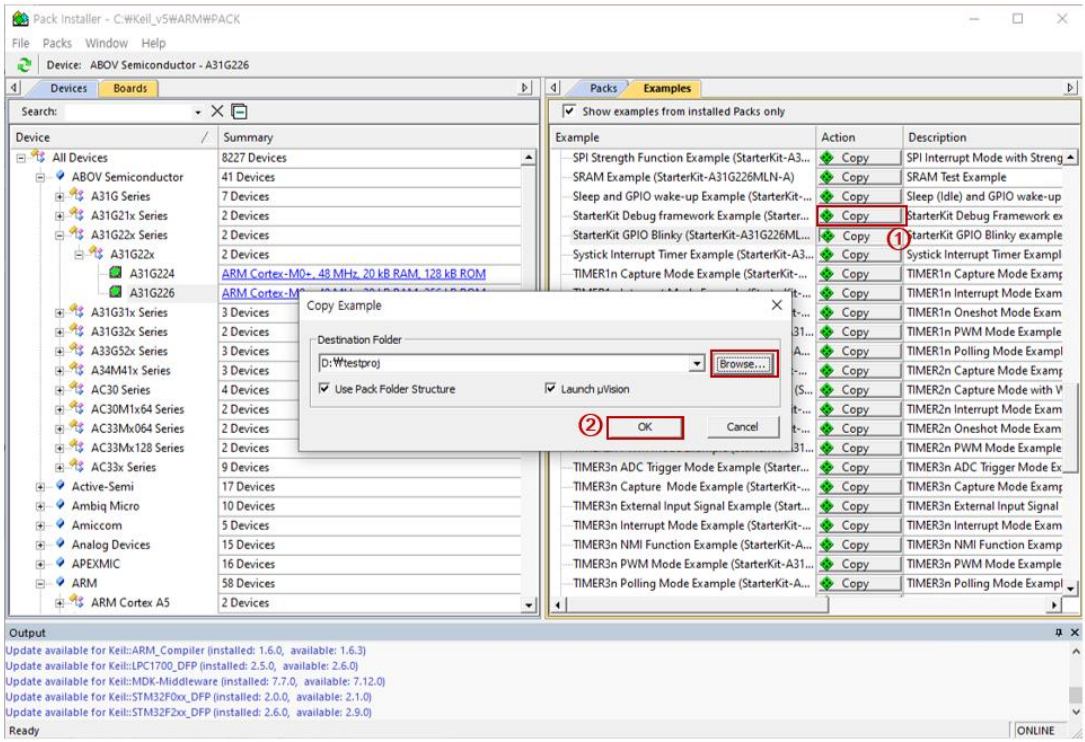


Figure 12. Create example project in Pack Installer

5. The example is copied to the destination directory and is displayed as shown below in uVision5.

① The destination folder contains a Keil project and the example files.

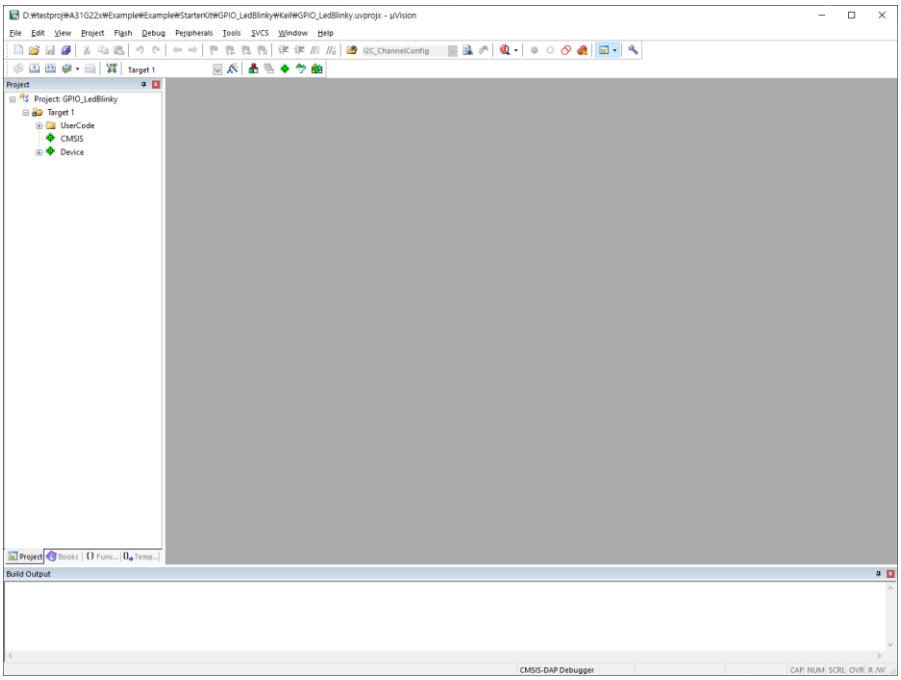


Figure 13. Example project created by the Pack installer

6. If using Example Code, you can download the Example_Code_[Device] software for your device from the ABOV Semiconductor website.

Once downloading is complete, unzip the file and run LED Blinky uVision5 in the example directory. See the Keil Setting Guide for detailed instructions.

■ **Documents**

Category	Title	Type	Language	Version	Issue Date	Size	Download
Datasheet	DS_A31G22x_Eng		Eng	1.01	2021-01-11	3.3 MB	
User's Manual	UM_A31G22x_Eng		Eng	1.01	2021-01-11	11.1 MB	
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Application Note	AN_InternalDataFlashMemoryGuide_A31G22x_Eng		Eng	1.00	2020-12-22	948 KB	
Application Note	AN_LEDCircuitDesignGuide_A31G22x_Eng		Eng	1.00	2020-12-22	1.1 MB	
Application Note	AN_OperatingModesandWake-upfunctionGuide_A31G22x_Eng		Eng	1.00	2020-12-22	529 KB	

Figure 14. Download the example from the ABOV homepage

3.5 Compiling LED_Blinky

Compile the created uVision5 LED Display Project.

① Select Rebuild to proceed with compilation.

② Compile result can be checked in Build Output. Make sure there are no errors in the project.

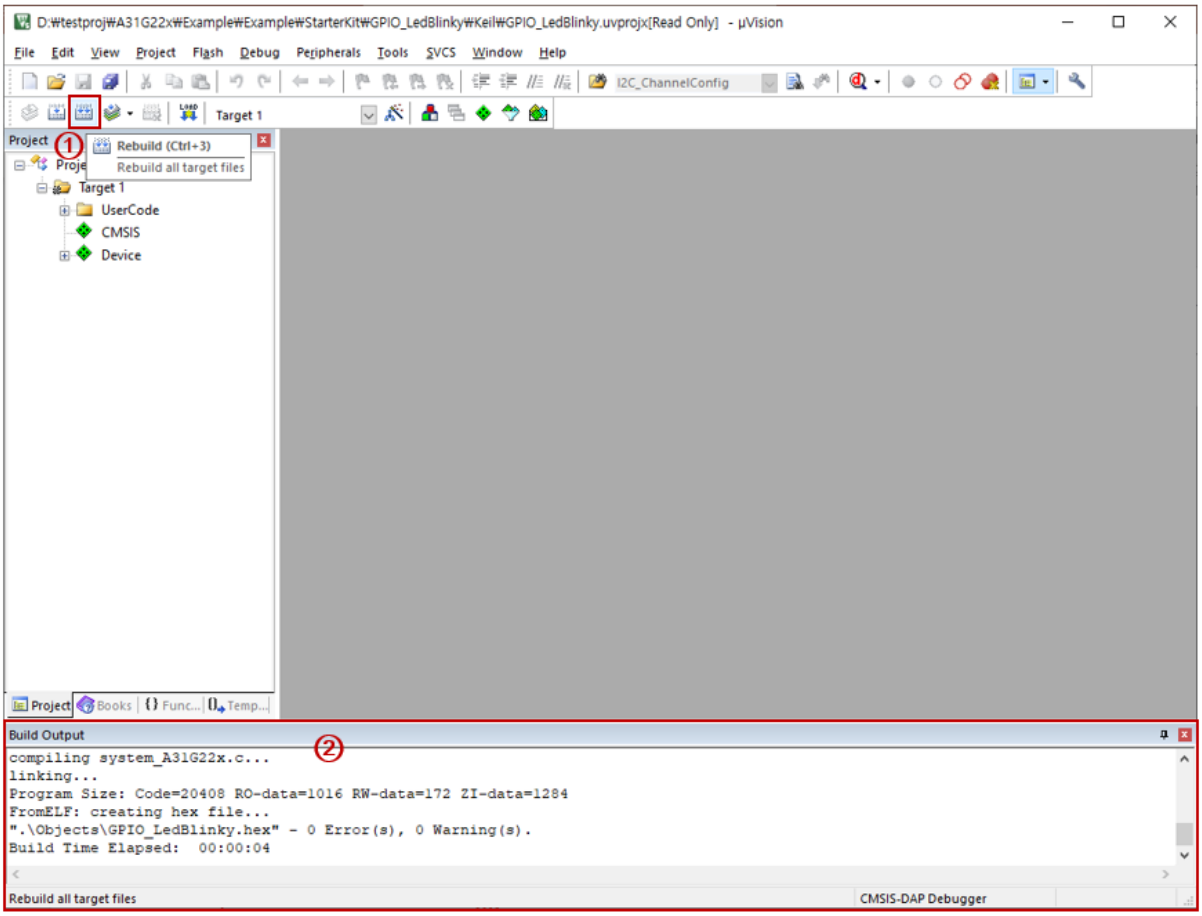


Figure 15. Compile result on LED Blinky Project

3.6 Download Led_Blinky

When compiling is completed without errors, you can begin programming the starter kit.

Downloading is enabled since the starter kit is connected to the USB port of the PC.

Configuration of Debugger

To download firmware to the target, A-Link debugger must be configured in options panel of the project.

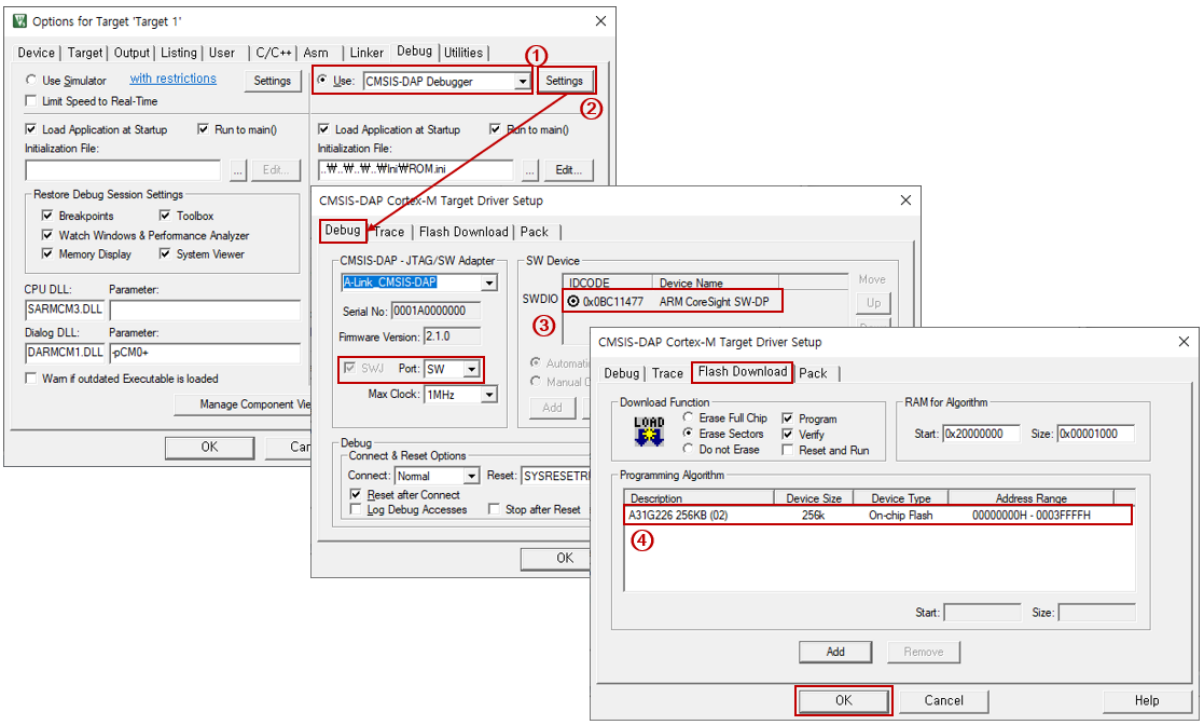


Figure 16. Debug settings for example project options

① Check that CMSIS-DAP Debugger is selected in the Debug tab. A-Link debug uses CMSIS-DAP Debugger Interface.

② Click Settings and check that A-Link CMSIS-DAP is selected in the Debug tab.

③ When A-Link debugger is well connected to target board, the target IDCODE and Device Name are displayed on the SWDIO of SW Device.

④ Check that the correct flash loaders of the device are displayed in the Flash Download tab.

Download Firmware

Click the download icon to download the firmware to the target and check that the download begins.

View the download results (2) in the Build Output pane.

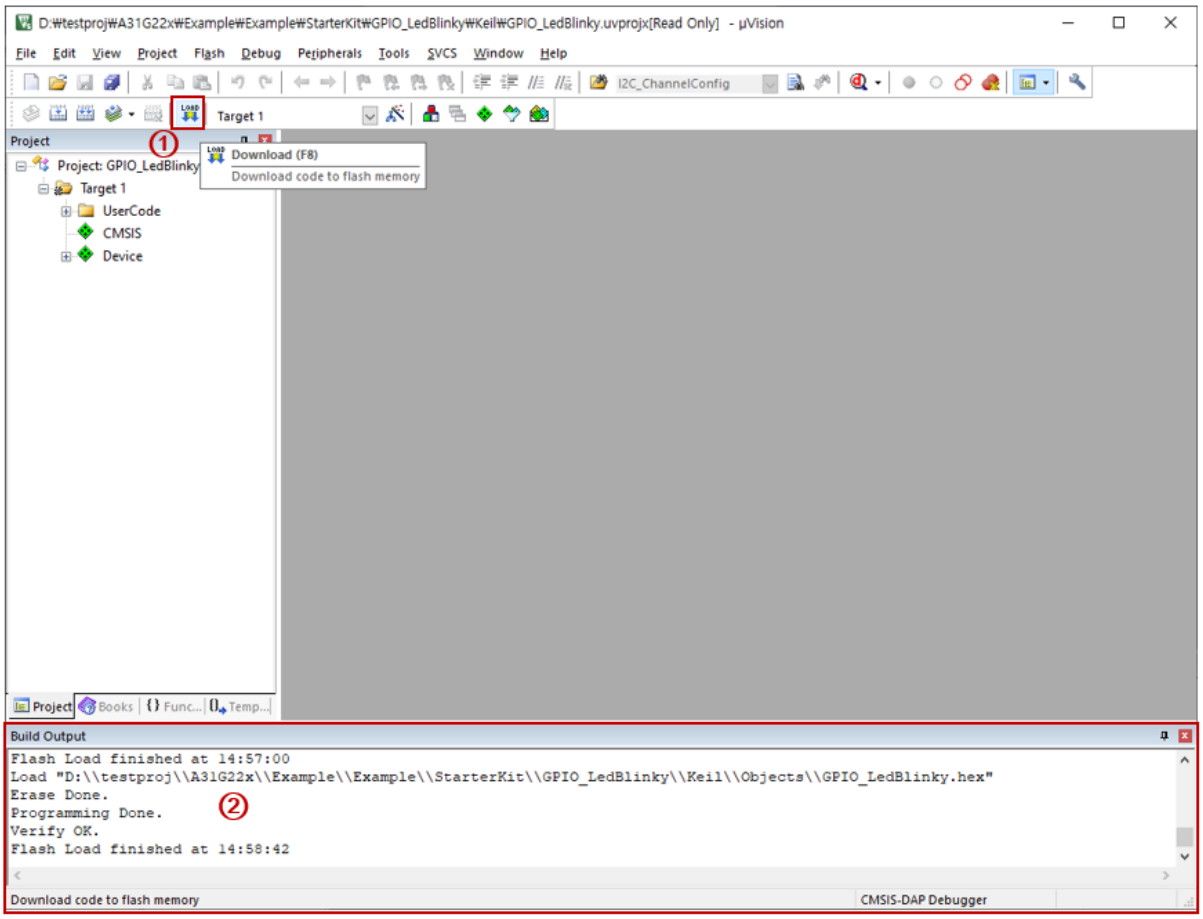


Figure 17. Firmware download and verification result

3.7 Debugging Led_Blinky

① Run ‘Start/Stop Debugger Session’ to enter debugger mode.

The starter kit must be connected for real-time, interactive debugging.

In debugger mode, the program can be executed per Run/Step/Stop Debugger Session.

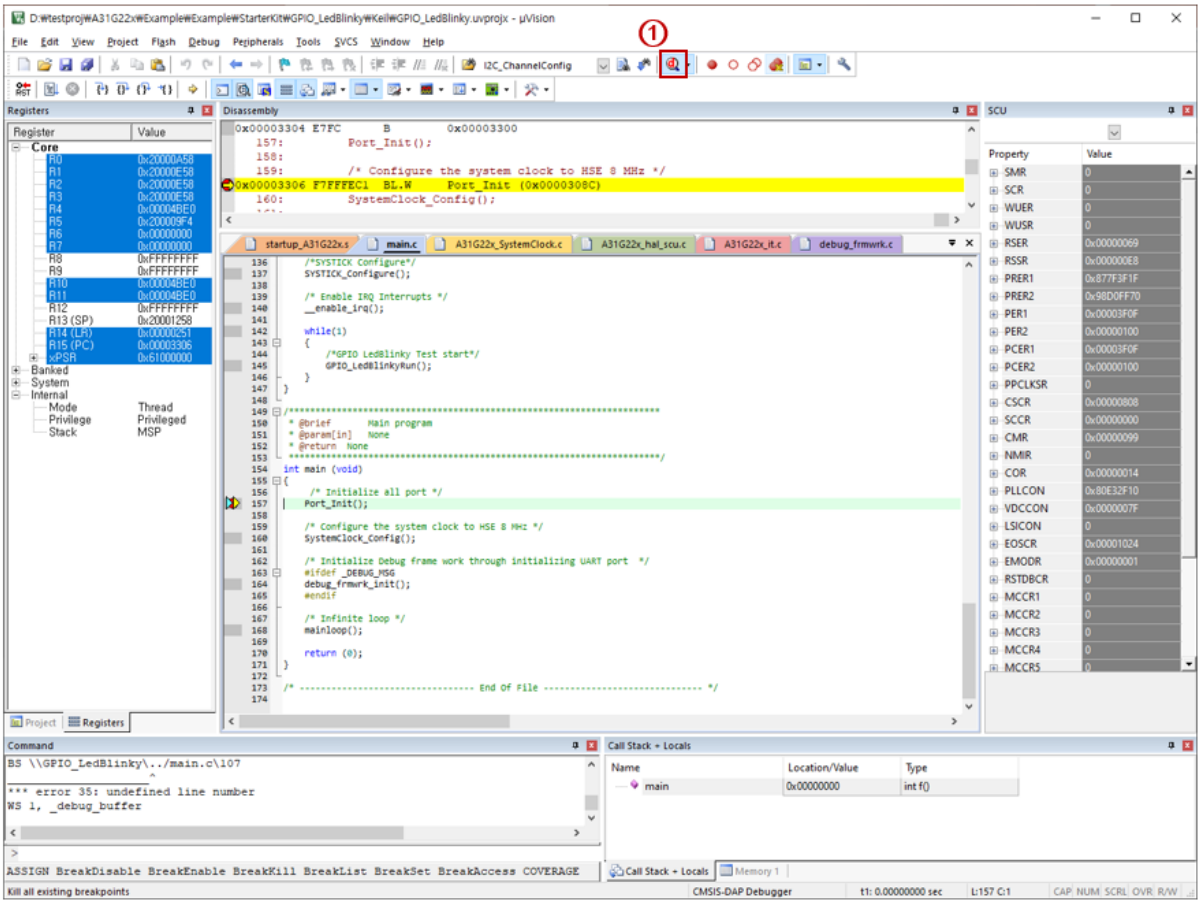


Figure 18. Debugging with ABOV A-Link and CMSIS-DAP

3.8 Check the Led_Blinky operation

How to check the LED operation

After normal download of the Led Blinky program, re-apply the power (removing the USB cable) to check whether the LED is on or off.

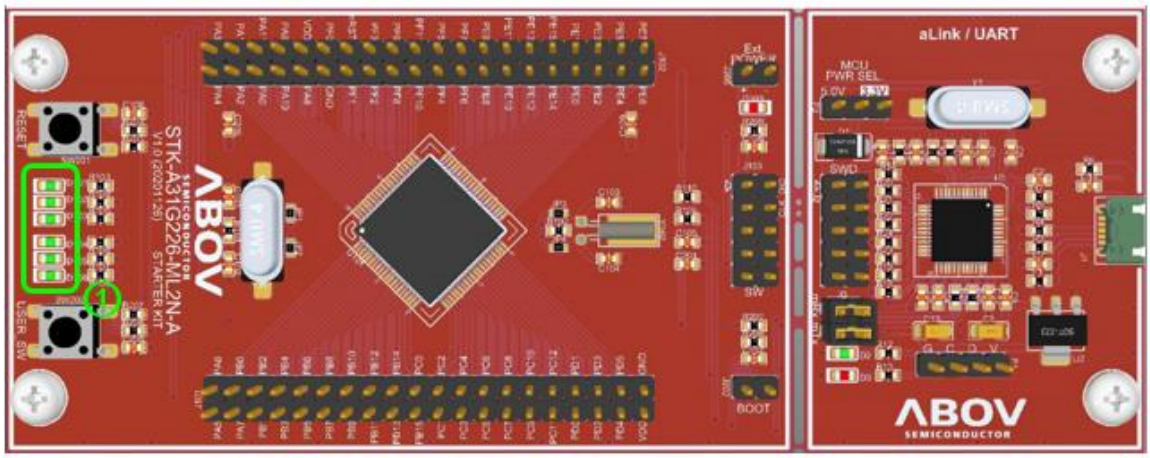


Figure 19. Check LED Operation

Revision history

Date	Revision	Description
Jun. 28, 2019	1.00	Document created.
Nov. 19, 2019	1.01	Add A31G324RLN, Updated website screen
Jan, 31, 2020	1.02	Add ABOV disclaimer
Feb. 5, 2020	1.03	Modify website's address
Nov. 2, 2020	1.04	Added targets – A31G226, A31G324
Jan. 21, 2021	1.05	Changed the template of document.
Oct. 30, 2021	1.06	Added target – A31L214
Oct. 17, 2022	1.07	Update the template of this document
Dec. 2, 2024	1.08	Updated the disclaimer.

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